Proto-Eight Dynamics (P8D): a small, testable model of how growth actually works【先天八卦動力學】

1) Start with a picture, not philosophy

Imagine two tanks with a pipe between them. One tank holds **capacity** (what you can reliably produce). The other holds **demand** (people who can actually buy and use what you offer). The height difference—**the potential gradient**—pushes water through the pipe. That flow is your **throughput**: orders shipped, revenue recognized, problems solved.

Most strategy debates are really about five questions:

1. Is the pipe actually there and open?
2. Is the product matched to what flows on the other side?
3. Do we keep the energy in the system long enough to compound?
4. Do our rules reduce friction or add it?
5. Do we monetize without choking the flow?

The **P8D model** answers these with a minimal set of variables and equations you can simulate in a spreadsheet.

2) The core equation of flow

Throughput at time t is:

$$y = \kappa_y \, \hat{e} \, m \, r \underbrace{\sqrt{s\,d}}_{\text{both sides matter}} \sigma\left(\frac{d-s}{\theta}\right)$$

Read it left to right:

- $\hat{e}$ (enablement) is high when rules create trust and low friction, or when **credit** substitutes for cash.
- m (match) is how well offer and need align.
- r (retention) keeps people, capital, and attention cycling instead of leaking away.
- $\sqrt{s\,d}$ says you need both capacity and demand; the scarcer side caps the flow.
- $\sigma\big((d-s)/\,\theta\big)$ opens the valve when demand exceeds capacity and closes it when you overbuild.

If you want only one mental model: *flow is fit times enablement times retention, gated by the bottleneck and the gradient.*

3) Where growth actually comes from

Capacity grows when you reinvest a portion of successful flow:

$$\dot{s} = \alpha_g \, g \, y - \mu_s s + J_{\text{innov}}$$

- g is **investment propensity** (how strongly success feeds capacity).
- J_{innov} captures discrete breakthroughs that "open new terrain" (a patent, a platform, a distribution deal).

Demand becomes *accessible* through reach and rules—sales, brand, distribution, and trustworthy processes:

$$\dot{d} = \beta_p \, p \, \hat{e} \, (1 - d) + \beta_y \, y - \mu_d d$$

- Raising **reach** p without lowering friction $\hat{e}$ is loud but not effective.
- Positive experience (throughput y) seeds network effects.

The **fit** m improves faster when you aren't wildly mismatched:

$$\dot{m} = \kappa_m \, [1 - |\, d - s \,|] \, (1 - m) - \mu_m m$$

When supply and demand are miles apart, customers teach you little; when they are close, each iteration adds a lot.

4) The two things that make or break compounding

First is **retention** r. It rises when buffers are healthy and when your "fun" has an aftertaste of self-control and care:

$$\dot{r} = \alpha_r \, (b - r) + \alpha_u \, u - \mu_r r$$

- Buffers b (cash/inventory) literally give time for second and third turns of the loop.
- u is the **quality of entertainment/engagement**—does it leave people more prosocial and focused, or just drained? The model rewards the former because it extends compounding.

Second is **buffers** themselves:

$$\dot{b} = \alpha_b \big(\pi_{\text{eff}} \, y \cdot \text{ARPU} - \zeta_b \, y \big) - \frac{1}{\tau_b} (b - b^{\star})$$

- π_{eff} improves with better fit and lower friction; **toll stations** (licenses/ads/data) raise ARPU without blocking the pipe.
- Don't starve the buffer: if outflows ζ_b (dividends, debt service) run ahead of earned flow, compounding stalls.

Together, **gradient + retention + buffer** are a necessary trio for sustained growth. Drop any one and momentum fizzles.

5) Rules: the most underrated growth lever

Rules can be "sand in the gears" or "lanes on the highway." In the model that is **friction** f vs **enablement** $\hat{e}$:

$$\dot{f} = -\gamma_L L + \text{shocks} - \mu_f (f - f^\star), \qquad \hat{e} = \text{clamp}(1 - f + \beta_k k, 0, 1)$$

- Good **standardization** L lowers friction every day (contracts, APIs, safety norms).
- In tight money conditions, **credit/clearing** k partly substitutes for cash to keep local loops alive.

6) What to measure on Monday morning

- **Match** m**:** win-rate in the ideal customer profile; funnel conversion; returns/complaints (inverse).
- **Enablement** $\hat{e}$**:** transaction-cost ratio, cycle time to close, credit availability.
- **Retention** r**:** employee/supplier churn; repeat-purchase rate; cohort decay.
- **Buffers** b**:** days of cash/inventory; working-capital turns; covenant headroom.
- **Throughput** y**:** fulfilled units or recognized revenue per period.
- **Toll density** n_{toll}**:** count of monetization nodes per process (licenses, data fees, ads, API calls).

7) How to use the model (playbook)

1. **Map your situation:** estimate s, d, m, r, b, f on [0,1] from KPIs.
2. **Simulate a quarter:** use the discrete update $\Delta x = \text{RHS} \cdot \Delta t$.
3. **Pick two levers for the next sprint:**
   - If $d \gg s$: invest g to lift s; protect b.
   - If $s \gg d$: raise p *and* L (reach with low friction), and improve m.
   - If r is low: pause flashy campaigns; rebuild buffers b and "good engagement" u.
   - If cash is tight: increase k (credit/clearing) to keep $\hat{e}$ high while you fix fundamentals.
4. **Insert toll stations** where the pipe already carries flow (licensing/ads/data), not where it blocks learning.
5. **Watch the warnings:** if recovery from small shocks slows, or variance in y spikes, deepen buffers, reduce friction, and pre-match before pushing reach.

8) Why this is "new" and useful

- It replaces metaphors with **one compact flow equation** and **six small updates** you can calibrate from ordinary KPIs.
- It unifies "product," "operations," and "governance" in one loop instead of separate dashboards.
- It makes the uncomfortable point explicit: **rules and buffers are growth levers**, not just compliance and cost.

Appendix: quick spreadsheet recipe

1. Put s, d, m, r, b, f in row 2 (0–1).
2. Compute $\Delta = d - s$, $\hat{e} = \max(0, \min(1, 1 - f + \beta_k k))$, $H = \sqrt{s\,d}$, $y = \kappa_y \hat{e} m r H \sigma(\Delta / \theta)$.
3. Update each state with Δt (e.g., 0.1):
    - $s \leftarrow s + (\alpha_g g y - \mu_s s + J)\Delta t$, etc., clamped to [0,1].
4. Repeat for 12 steps; track y and warnings (variance ↑, recovery ↓).
5. Sensitivity-test by toggling one lever at a time (raise L, then p, then g, etc.).

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I am merely a midwife of knowledge.